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Image restoration model compression via mamba-oriented heterogeneous knowledge distillation

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ABSTRACT

The transformer-based Image Restoration (IR) models usually require heavy computation and intensive parameters. To improve the efficiency of IR models, we propose a knowledge distillation strategy for image restoration model compression, specifically instantiated as Mamba-oriented Heterogeneous Knowledge Distillation (MHKD) or the Transformer-to-Mamba architecture pair. Specifically, we firstly pre-train a cumbersome yet high-quality IR model with transformer blocks as the teacher network. Secondly, we construct a lightweight IR model with the vision Mamba blocks as the student network. Subsequently, MHKD is used to transfer the heterogeneous knowledge from the teacher network to the student network. The key elements of MHKD are the feature filter module and feature interface module. The former is designed to reduce the redundant information from the teacher network, and the latter aims to translate the heterogeneous feature of teacher network for the student network with the designed bi-direction knowledge transferring loss. Extensive experiments across multiple IR benchmarks confirm that, for the Transformer-to-Mamba distillation setting, MHKD can assist a Mamba-based IR model with compact size ($\approx 716\text{K}$ parameters) to achieve comparable or even better performance than state-of-the-art models.

Image Restoration (IR) aims to restore a clean image from its contaminated version caused by degradation factors, e.g., noise, blurriness, low illumination. As a fundamental computer vision task, IR is critical for a wide range of realistic applications, such as video surveillance, autonomous driving. Because of existing infinite feasible solutions, IR is a severely ill-posed problem (Sun et al., 2025; Yang et al., 2024; Zhang et al., 2025c). Therefore, convention methods (Li et al., 2016) endeavored to constrain the solution space through exploring various hand-crafted priors based on the statistical observations, lacking robustness for the complex scenes. In contrast, deep learning based methods (Cui et al., 2024; Gao et al., 2025; Zhang et al., 2024a, 2025a, 2024b) implicitly acquire general priors from the abundant data, which have become prevailing with satisfactory results. Moreover, IR has also played a pivotal role in advancing underwater image analysis (Li et al., 2025b; Wang et al., 2023, 2025; Xu et al., 2025a,b). Recent advances leverage multi-layer information fusion and self-organized stitching (Sun & Tian, 2023; Wang et al., 2024a), significantly boosting performance on diverse underwater scenes.

Currently, to unify diverse restoration tasks, developing a general-purpose model with advanced neural architectures is a hot research

topic. Also, the dominate neural architecture in IR community has shifted from Convolutional Neural Networks (CNNs) (Bai et al., 2024; Ren et al., 2019; Zhang et al., 2025b) to transformers (Ma et al., 2025; Yu et al., 2024; Zheng et al., 2024). As shown in Fig. 1(a), the essential component of transformers is the multi-head self-attention, which wins advantages of long-range dependency modeling and efficient parallel computing, yet incurring a quadratic complexity with respect to the input size, making the transformer computationally expensive and memory intensive when applying to high-resolution images and limited resource applications. Motivated by these problems, a compelling set of neural architectures (Beck et al., 2025; Peng et al., 2023) have sprung up to challenge transformer in natural language processing and subsequently extended to computer vision. Notably, Mamba (Dao & Gu, 2024; Zhu et al., 2024) behaves very competitive with the transformer, yet having linear time and constant memory complexity with respect to the input size. As shown in Fig. 1(b), the fundamental principle of Mamba is the state space model.

As a low-level computer vision task, it is highly demanding to develop lightweight yet swift models for IR tasks. Driven by this, Li et al. (2020) pioneered to exploit Knowledge Distillation (KD) to implement

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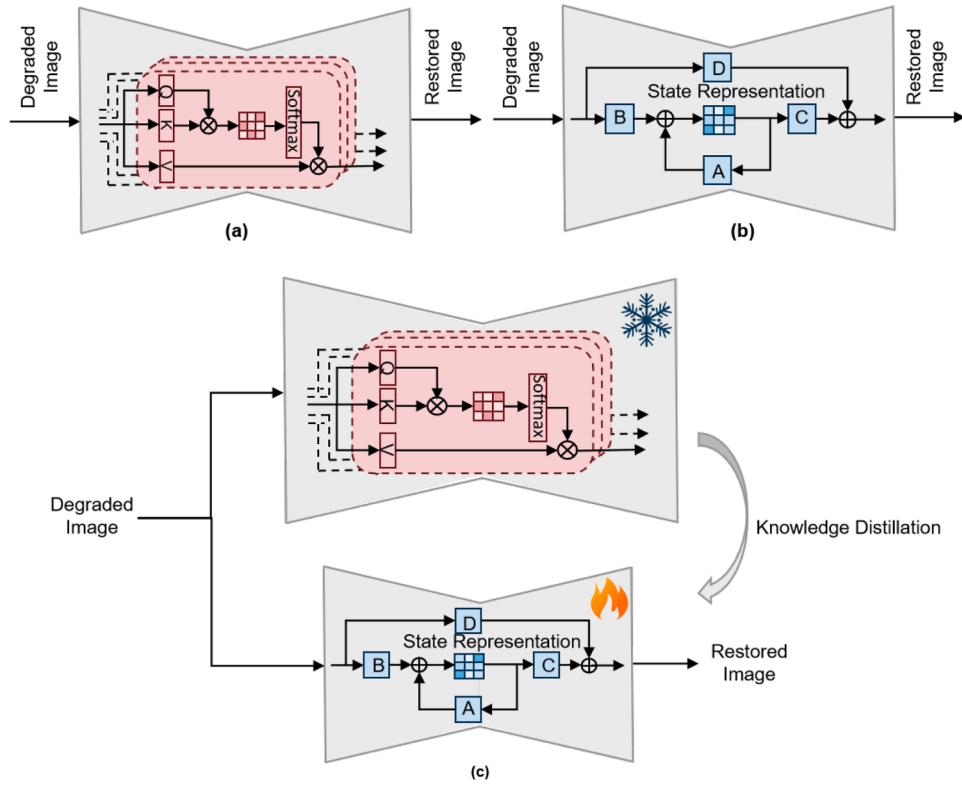


Fig. 1. (a) Transformer-based model. It can establish long-range dependency with the multi-head self-attention, yet incurring a quadratic complexity with respect to input size. (b) Mamba-based model. It employs the state space model to have linear time complexity with respect to the input size. (c) Our method. It takes the transformer-based model and the Mamba-based model as the teacher network and student network, respectively. Knowledge distillation is used to implement the model compression, thus approaching a lightweight yet fast image restoration model.

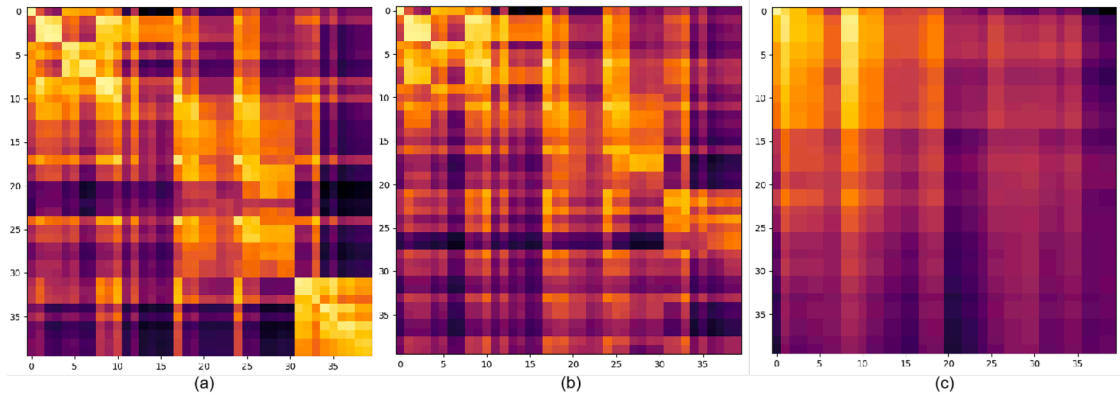


Fig. 2. Centered Kernel Alignment (CKA) analysis. (a) Transformer vs. Transformer. (b) Mamba vs. Mamba. (c) Transformer vs. Mamba. Wherein, (a) and (b) demonstrate feature similarity between homogeneous models, (c) demonstrates feature dissimilarity between heterogeneous models.

model compression. However, their work only focused on a sub-task of image super-resolution, and lacks a formal formulation to flourish the research in this line. Further, considering the aforementioned complementary strengths of transformer and Mamba, we formally propose a new setting of knowledge distillation-based image restoration model compression and illustrate its basic formulation in Fig. 1(c). Generally, it can be decomposed into three main parts: (1) Pre-training a teacher network, that is a cumbersome yet high-quality IR model built upon the transformer. (2) Constructing a student network, that is a lightweight IR model based on Mamba. (3) Designing a specialized KD method to shift the knowledge from the teacher network to the student network.

Moreover, the key of the proposed formulation lies in how to design the specialized KD method, being significantly challenged by the

representation gap between the heterogeneous architectures of transformer and Mamba. To support this, we adopt Centered Kernel Alignment (CKA) (Kornblith et al., 2019) to compare features extracted by transformer-based and Mamba-based models. The resulted feature similarity heatmaps are shown in Fig. 2, where homogeneous architectures tend to learn similar features at layers of similar positions in Fig. 2(a) and (b), while heterogeneous architectures are remarkably different in Fig. 2(c). Consequently, it is detrimental for the compression performance to directly match the distant features from these two homogeneous architectures. To circumvent this problem, we propose the Mamba-oriented Heterogeneous Knowledge Distillation (MHKD). Specifically, MHKD designs the feature filter module, which takes the auto-encoder to apply the self-reconstruction, thus preserving the crit-

ical feature information from the teacher network in the encoder part. Subsequently, MHKD designs the feature interface module, which uses the bi-direction knowledge transferring loss to transfer the above critical feature better for Mamba-based model.

The proposed feature filtering and bidirectional interface alignment mechanisms theoretically mitigate representational mismatch between Transformer and Mamba architectures by addressing the root causes of feature discrepancy. The feature filter module, structured as an autoencoder, compresses the teacher's Transformer-derived features to retain only task-critical, architecture-agnostic information while discarding attention-specific redundancy and noise, which directly reduces the raw distribution gap between teacher and student features before alignment. The bidirectional interface module further resolves structural inconsistency by mutually projecting filtered teacher features into the Mamba state-space and student features into the Transformer attention space, establishing a shared latent translation pathway that enables compatible knowledge transfer rather than forcing direct alignment of inherently dissimilar representations. Together, these two components form a unified conceptual framework that first cleanses misaligned information and then enables reciprocal feature adaptation, theoretically ensuring smoother and more robust knowledge transfer across heterogeneous architectures.

To sum up, the main contributions are:

- We introduce a knowledge distillation-based strategy for image restoration model compression and specify its formulation, focusing on the specific case where a Transformer teacher compresses a Mamba student. This instantiation effectively integrates the complementary strengths of Transformer and Mamba within a distillation framework.
- We propose a new Mamba-oriented Heterogeneous Knowledge Distillation (MHKD) method, which is customized to realize information interaction between the heterogeneous neural architectures of transformer and Mamba, result in beneficial distillation performance.
- We conduct extensive experiments across numerous IR tasks to showcase the effectiveness of our method.

The remainder of this paper is organized as follows. In Section 2, we mainly review the related works and discuss their advantages and disadvantages. In Section 3, we formulate the proposed algorithm from the framework to the details and provide the pseudo code for sketching the whole proposed algorithm conveniently. In Section 4, we describe the extensive experimental results and comparisons with state-of-the-art methods. Finally, we present a conclusion with some future research directions in Section 5.

1. Related works

1.1. Image restoration (IR)

Benefiting from the outstanding non-linear representations, deep learning-based methods have become the mainstream in IR community. These methods use neural architectures to directly learn the mapping from degraded images to clean counterparts. The early popular architectures are CNNs, which are applied into each specific IR task, such as AECR-Net (Wu et al., 2021) for dehazing, PReNet (Ren et al., 2019) for deraining, MIMO-UNet (Cho et al., 2021) for deblurring, yet suffering the local receptive field issue. Subsequently, the powerful transformers with long-range dependency modeling capability are introduced into the IR community, leading to models that aim to unify multiple IR tasks. In this context, numerous remarkable models, such as SwinIR (Liang et al., 2021), Uformer (Wang et al., 2022), Restormer (Zamir et al., 2022) have been successively proposed. However, the transformer-based models are highly challenged by the computational complexity of self-attention. Although a series of delicate attentions (Chen et al., 2022; Zhou et al., 2024) have been designed, the dilemma of the trade-off between efficient computation and global modeling is still not essentially resolved.

Recently, the emergent architecture of Mamba exhibits promising performance with linear complexity. Mamba is also introduced into IR, exemplified with MambaIR (Guo et al., 2024), VmambaIR (Shi et al., 2025). The former added image-specific inductive biases with a 3×3 depth-wise convolution and squeeze-and-excitation channel-attention module to overcome Mamba's weaknesses at the pixel level, while the latter incorporated Omni selective scan mechanism to aggregate context from its complete neighborhood. Furthermore, MambaIRv2 (Guo et al., 2025) introduced an attentive state-space module that injected learnable prompts to give every pixel a full-image receptive field in one linear pass. MaIR (Li et al., 2025a) proposed nested S-shaped scanning strategy and sequence shuffle attention block to keep the local relationships and spatial continuity. Currently, researchers are striving to integrate the complementary strengths of transformer and Mamba in many fields (Fei et al., 2024; Lieber et al., 2024). Following this line, we explore the integration of Transformer and Mamba in the IR domain using knowledge distillation, specifically in a teacher-student compression setup.

Driven by the realistic demands, constant efforts have been devoted to efficient IR models with low capacity and fast computation. For instance, Choi et al. (2024) designed a lightweight network through building efficient convolutions upon MobileNet variant. Besides, some popular model compression techniques are adopted into IR. For example, Zhang et al. (2021) implemented structure-regularized pruning for the Super-Resolution (SR) model. Shi et al. (2023) proposed memory-oriented pruning to reduce the structural redundancy of IR models. Li et al. (2020) quantized SR models and then applied KD to maintain performance. Differently, we target the entire IR field and propose a formal formulation of knowledge distillation-based image restoration model compression, which is realized by employing the complementary advantages of transformer and Mamba.

1.2. Model compression via knowledge distillation

Knowledge Distillation (KD) has emerged as an effective model compression technique, which transfers the knowledge from a large well-performing teacher network to a lightweight student. The traditional KD methods largely lean toward image classification, being broadly divided into output-based and feature-based distillation. The former attempts to match the predicted outputs between the teacher and student, such as DKD (Zhao et al., 2022), MLKD (Jin et al., 2023), LS (Sun et al., 2024), while the latter aims to align the intermediate features between the teacher and student, such as FitNets (Romero et al., 2015), CAT (Guo et al., 2023), V_k D (Miles et al., 2024). KD methods are also designed to tailor for other high-level computer vision tasks. For example, Yang et al. (2022) employed contextual pixel affinity as the distillation knowledge for the semantic segmentation. Chang et al. (2023) implemented multiple KD methods for the DETR-style detectors in the aspects of logits, features, and convergence rate. Albeit significant progress with above KD methods, they are elaborately designed for high-level computation vision. How to adapt KD to the low-level computation vision is worth further studying.

With the continuous emergence of diverse neural architectures, homogeneous knowledge distillation has received increasing attention in recent years. For example, Touvron et al. (2021) successfully trained a student with transformer under the supervision of a teacher with CNNs in image classification task. Wang et al. (2024b) implemented homogeneous distillation between DETR and CNN detector through constructing coordination-based distillation points. Hao et al. (2024) projected the intermediate features into an aligned latent space, thus realizing free distillation across any homogeneous architectures of transformers, CNNs and MLPs. Wu et al. (2024) extracted the low-frequency components of features to mitigate the representational disparity between the transformers and CNNs. Unlike existing heterogeneous distillation methods designed for high-level vision and CNN-Transformer alignment, our method is the first framework tailored for low-level image restoration to bridge Transformer and Mamba architectures. Furthermore, in con-

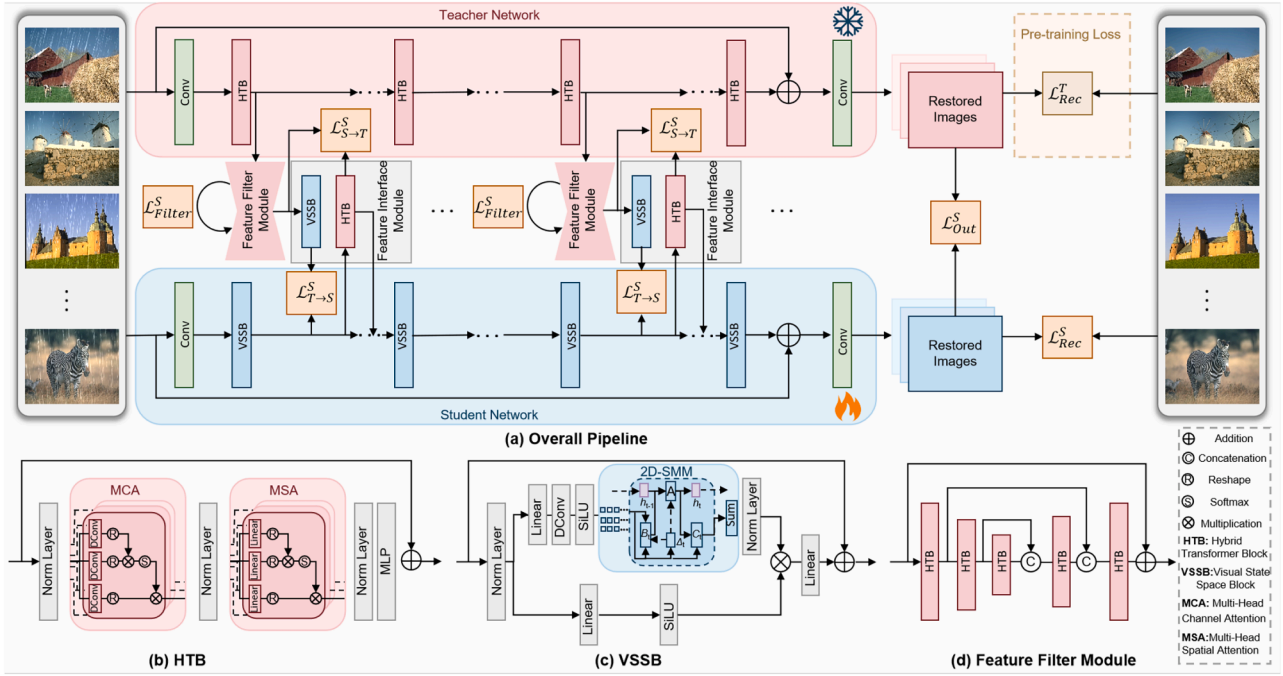


Fig. 3. (a) Overall Pipeline of the proposed Mamba-oriented Heterogeneous Knowledge Distillation (MHKD). MHKD designs the Feature Filter Module (FFM) and Feature Interface Module (FIM) to efficiently implement heterogeneous knowledge distillation between the transformer-based IR model and Mamba-based IR model. (b) HTB. It mainly consists of Multi-head Channel-wise Attention (MCA) and Multi-head Spatial-wise Attention (MSA). (c) VSSB. It employs the 2D selective scan module to construct the Mamba-based IR block. (d) Feature Filter Module. It is an autoencoder with transformer layers.

trast to generic unidirectional feature projection, our methods uses a Feature Filter Module to reduce teacher redundancy and a bidirectional Feature Interface Module for stable cross-architecture knowledge transfer. These dedicated designs enable effective distillation from heavy Transformer teachers to tiny Mamba students, delivering a lightweight model with performance and efficiency beyond prior cross-architecture solutions.

In summary, this work is the first to formally define a knowledge distillation framework for general image restoration (IR) model compression, moving beyond prior task-specific or architecture-homogeneous methods. It uniquely addresses the significant feature gap between a Transformer teacher and a Mamba student via a novel Feature Filter Module and bidirectional Feature Interface Module, enabling effective heterogeneous distillation. Consequently, it achieves superior efficiency, compressing a high-performance model into an extremely lightweight student that remains competitive across multiple IR tasks.

2. The proposed method

2.1. Overall pipeline

To attain an efficient IR model via heterogeneous distillation, we propose Mamba-oriented Heterogeneous Knowledge Distillation (MHKD), which compresses a Transformer teacher into a Mamba student. The overall pipeline of MHKD is illustrated in Fig. 3. Generally, MHKD consists of three stages, i.e., (1) Teacher network with transformer. This stage firstly employs the transformer to build a cumbersome yet high-quality IR model, and then pre-trains this model as the teacher network. (2) Student network with Mamba. This stage aims to construct a lightweight IR model based on Mamba as the student network. (3) Model Compression. This stage implements knowledge transferring from the teacher to the student via MHKD, which holds core components of the feature filter module and the feature interface module. The above three stages will be detailed in the following sections.

2.2. Teacher network with transformer

The teacher network mainly consists of three modules of shallow feature extraction, deep feature extraction, and image reconstruction, denoted as T . Given an input image I_i , which is implemented with a 3×3 convolutional layer, resulting in the shallow feature $F_0^T \in \mathcal{R}^{H \times W \times C}$, where H , W and C mean the height, width and channel numbers. The deep feature extraction module is composed of encoder E^T and decoder D^T , which can be decomposed into 4 stages. Assume the output feature from the encoder or decoder in any stage is denoted as F_j^T . The stages in E^T are stacked with numerous Hybrid Transformer Block (HTB) and a down-sampling layer, while the ones in D^T are stacked with numerous HTB and a up-sampling layer. The block numbers in each stage are respectively denoted as $N_1^T, N_2^T, N_3^T, N_4^T$. The parity stage of the encoder and decoder are linked by the skip connection. The feature from the last stage of D^T can be denoted as $F_d^T \in \mathcal{R}^{H \times W \times C}$. Finally, a 3×3 convolutional layer is applied on F_d^T to generate residual image $I_r^T \in \mathcal{R}^{H \times W \times 3}$. Adding I_r^T and I_i together produces the output image $I_1^T = I_r^T + I_i \in \mathcal{R}^{H \times W \times 3}$.

In addition, HTB is produced by combing Multi-head Channel-wise Attention (MCA) and Multi-head Spatial-wise Attention (MSA) together. Given the input feature of $Z_{in}^T \in \mathcal{R}^{H \times W \times C}$, the transformation in HTB is:

$$\begin{aligned} Z_1^T &= Z_{in}^T + MCA(LN(Z_{in}^T)), \\ Z_2^T &= Z_1^T + MLP(LN(Z_1^T)), \\ Z_3^T &= Z_2^T + MSA(LN(Z_2^T)), \\ Z_{out}^T &= Z_3^T + MLP(LN(Z_3^T)), \end{aligned} \quad (1)$$

where, Z_{out}^T is the output feature of HTB. LN and MLP stand for normalization layer and fully connected layers, respectively. The heads number of attention are set to be H . In MCA, the channels of Z_{in}^T are divided into G parts, with each is denoted as Z_{in}^g , the self-attention is

calculated as:

$$\mathbf{Z}_{in}^g = \text{Softmax}\left\{\frac{(\mathbf{Z}_{in}^g \mathbf{W}_Q^h) \cdot (\mathbf{Z}_{in}^g \mathbf{W}_K^h)^T}{\sqrt{d_1}}\right\} \cdot (\mathbf{Z}_{in}^g \mathbf{W}_V^h), \quad (2)$$

where, $\mathbf{W}_Q^h, \mathbf{W}_K^h, \mathbf{W}_V^h$ are the depth-wise separable convolution. d_1 is the channel dimension in each head. In MSA, $\mathbf{Z}_2^T \in \mathcal{R}^{H \times W \times H}$ is partitioned into non-overlapping local windows in size of $M \times M$, with each is denoted as $\mathbf{Z}''_w \in \mathcal{R}^{M \times M \times C}$, the self-attention is calculated as:

$$\mathbf{Z}''_w = \text{Softmax}\left\{\frac{(\mathbf{Z}''_w \mathbf{P}_Q^h) \cdot (\mathbf{Z}''_w \mathbf{P}_K^h)^T}{\sqrt{d_2}}\right\} \cdot (\mathbf{Z}''_w \mathbf{P}_V^h), \quad (3)$$

where, $\mathbf{P}_Q^h, \mathbf{P}_K^h, \mathbf{P}_V^h$ are linear layers. d_2 is the channel dimension in each head. The above attention operations are performed for H times in parallel, and the outputs are reshaped and merged together.

After building the teacher network, we sample a batch of paired images from the dataset, where any degraded image and its corresponding clean image are denoted as I_i, y_i , respectively. I_i is input into T , yielding the output restored image $\hat{I}_i^T$. The reconstruction loss between the restored image $\hat{I}_i$ and the ground truth y_i is formulated as:

$$\mathcal{L}_{Rec}^T = \sum_{i=1}^N |\hat{I}_i^T - y_i|_1, \quad (4)$$

where, N is the batch size, $|\cdot|_1$ is the l_1 distance. The pre-training of the teacher is accomplished upon $\mathcal{L}_{Rec}^T$.

2.3. Lightweight student network with mamba

The student network generally follow the same structure with the teacher network, comprising of primary modules of (1) shallow feature extraction, (2) deep feature extraction, and (3) image reconstruction. We denote the student network as S . Similar to the teacher network, the student network also employs the 3×3 convolutional layer in the first and the third module, and the second module also consists of the encoder and decoder with 4 stages. The output feature from any stage is denoted as $\mathbf{F}_j^S$. The deep feature extraction model is stacked by multiple Residual State-Space Groups (RSSG). And each RSSG contains several Vision State Space Blocks (VSSB) and an additional convolution layer. Given an input image $\mathbf{I}_i \in \mathcal{R}^{H \times W \times 3}$, the output of the first module is denoted as $\mathbf{F}_0^S \in \mathcal{R}^{H \times W \times C}$, which is further processed by the second module, yielding the deep feature $\mathbf{F}_d^S \in \mathcal{R}^{H \times W \times C}$. With $\mathbf{F}_d^S \in \mathcal{R}^{H \times W \times C}$, the third module generates the residual image $\mathbf{I}_r^S \in \mathcal{R}^{H \times W \times 3}$. Adding $\mathbf{I}_r^S$ and $\mathbf{I}_i$ together produces the output image $\hat{\mathbf{I}}_i^S \in \mathcal{R}^{H \times W \times 3}$.

Additionally, given the input feature $\mathbf{Z}_{in}^S \in \mathcal{R}^{H \times W \times C}$, VSSB implements the following transformation:

$$\begin{aligned} \mathbf{Z}_1^S &= LN(\mathbf{Z}_{in}^S), \\ \mathbf{Z}_2^S &= LN(2D-SMM(SiLU(DWconv(Linear(\mathbf{Z}_1^S))))), \\ \mathbf{Z}_3^S &= SiLU(Linear(\mathbf{Z}_1^S)), \\ \mathbf{Z}_4^S &= Linear(\mathbf{Z}_2^S \odot \mathbf{Z}_3^S), \\ \mathbf{Z}_{out}^S &= \mathbf{Z}_{in}^S + \mathbf{Z}_4^S, \end{aligned} \quad (5)$$

where, LN , $Linear$ stand for normalization layer and linear layer. $SiLU$ and $DWConv$ are the activation function and depth-wise convolution. $\odot$ denotes the Hadamard product. $2D-SMM$ represents 2D selective scan module.

2.4. Mamba-oriented heterogeneous knowledge distillation (MHKD)

The core components of MHKD are Feature Filter Module (FFM) and Feature Interface Module (FIM). The design motivation for the FFM and FIM stems from the significant feature representation gap between the Transformer-based teacher and Mamba-based student, as evidenced by the Centered Kernel Alignment (CKA) analysis in Fig. 2. The FFM acts as an autoencoder to compress and filter the teacher's features, extracting

a more generalized, architecture-agnostic representation by reconstructing the input features through a bottleneck—this theoretically aligns with minimizing information redundancy. The FIM project the teacher's filtered features into the Mamba space and a tiny HTB to project the student's features into the Transformer space, enforcing alignment in a shared latent space through reversible transformation. As shown in Fig. 3(a), FFM's encoder-decoder uses lightweight Transformer layers, while the FIM employs single-block VSSB and HTB modules to minimize computational overhead during distillation.

Given a batch of N images sampled from the dataset, i.e., $\{(\mathbf{I}_i, \mathbf{y}_i)\}_{i=1}^N$, where $(\mathbf{I}_i, \mathbf{y}_i)$ is the i th degraded image and its corresponding ground truth. Feed $\mathbf{I}_i$ into S , producing the feature in each stage of $\mathbf{F}_j^S$ and the output restored image $\hat{I}_i^S$. l_1 distance is adopted to establish the following reconstruction loss for the student network:

$$\mathcal{L}_{Rec}^S = \sum_{i=1}^N |\hat{I}_i^S - \mathbf{y}_i|_1. \quad (6)$$

The parameters in the teacher network are frozen after pre-training. Meantime, $\mathbf{I}_i$ is fed into T , producing the feature in each stage of $\mathbf{F}_j^T$ and the output restored image $\hat{I}_i^T$. During the KD process, the student is firstly encouraged to learn the output-based knowledge from the teacher. Thus, the output-based distillation loss is constructed as :

$$\mathcal{L}_{Out}^S = \sum_{i=1}^N |\hat{I}_i^S - \hat{I}_i^T|_1. \quad (7)$$

Besides, feature maps of the teacher are valuable information to assist the student to gain better performance. As mentioned earlier, feature discrepancy are observed between the heterogeneous neural architectures of transformer-based model and Mamba-based model. Therefore, we firstly design a Feature Filter Module (FFM) to reduce the redundant information from the transformer-based teacher network. FFM adopts the form of the autoencoder to reconstruct its own inputs, thus capturing the hidden core feature information. FFM consists of the encoder and decoder with the transformer layers, which act as the mapping function $f(\cdot)$ and $g(\cdot)$, respectively. Given the feature maps $\mathbf{F}_j^T$, the encoder maps it into the hidden space to get the core feature $\mathbf{H}_j^T$, and the decoder map the hidden feature $\mathbf{H}_j^T$ to reconstruct the input. Mathematically, the above process is described as:

$$\begin{aligned} f : \mathbf{F}_j^T &\rightarrow \mathbf{H}_j^T, \\ g : \mathbf{H}_j^T &\rightarrow \mathbf{F}_j^T, \\ \mathcal{L}_{Filter}^S &= \sum_{i=1}^N \sum_{j=1}^4 |\mathbf{F}_{ij}^T - g[f(\mathbf{F}_{ij}^T)]|_1. \end{aligned} \quad (8)$$

Moreover, we also design a Feature Interface Module (FIM) to translate the heterogeneous feature of teacher for the student. FIM firstly applies the VSSB to change the feature from the teacher into Mamba-based feature, and establishes the one-way feature distillation loss from the teacher to the student, denoted as $\mathcal{L}_{T \rightarrow S}^S$. Meantime, FIM inserts a tiny HTB into the student network with the aim of changing the feature from the student network into transform-based feature, whilst constructing the one-way feature distillation loss from the student to the teacher, denoted as $\mathcal{L}_{S \rightarrow T}^S$. Then we construct the bi-direction knowledge transferring loss by linearly combining $\mathcal{L}_{T \rightarrow S}^S$ and $\mathcal{L}_{S \rightarrow T}^S$, which is:

$$\begin{aligned} \hat{\mathbf{H}}_j^T &= VSSB(\mathbf{H}_j^T) \\ \hat{\mathbf{F}}_j^S &= HTB(\mathbf{F}_j^S) \\ \mathcal{L}_{T \rightarrow S}^S &= \sum_{i=1}^N \sum_{j=1}^4 |\hat{\mathbf{H}}_j^T - \mathbf{F}_j^S|_1, \\ \mathcal{L}_{S \rightarrow T}^S &= \sum_{i=1}^N \sum_{j=1}^4 |\hat{\mathbf{F}}_j^S - \mathbf{H}_j^T|_1, \\ \mathcal{L}_{Bi}^S &= \mathcal{L}_{T \rightarrow S}^S + \alpha \mathcal{L}_{S \rightarrow T}^S. \end{aligned} \quad (9)$$

Table 1

Quantitative comparison results on denoising task, where last two columns list the parameters and Flops of each method. σ means corruption levels of Gaussian noise. Our MHKD student achieves competitive results with only 716K parameters, significantly fewer than state-of-the-art models.

Method	BSD68						Urban100						#Param↓	#Flops↓
	$\sigma = 15$		$\sigma = 25$		$\sigma = 50$		$\sigma = 15$		$\sigma = 25$		$\sigma = 50$			
	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑		
SwinIR (Liang et al., 2021)	34.04	0.9341	31.35	0.8898	27.80	0.7862	34.30	0.9472	31.78	0.9172	27.86	0.8431	3.48M	53.25G
Uformer (Wang et al., 2022)	33.86	0.9372	31.18	0.8948	27.82	0.8060	34.25	0.9514	31.93	0.9267	28.46	0.8715	52.66M	121.77G
Restormer (Zamir et al., 2022)	34.05	0.9349	31.44	0.8923	28.26	0.8080	34.35	0.9537	31.91	0.9274	28.63	0.8720	26.88M	40.16G
Stoformer (Xiao et al., 2022)	33.86	0.9372	31.17	0.8947	27.84	0.8078	34.27	0.9515	31.79	0.9248	28.60	0.8745	50.47M	123.97G
CAT (Chen et al., 2022)	33.87	0.9319	31.28	0.8877	28.19	0.8026	34.15	0.9511	31.70	0.9157	29.01	0.8682	26.64M	39.44G
ART (Jiale et al., 2023)	33.85	0.9324	31.23	0.8872	28.13	0.8032	34.21	0.9480	31.49	0.9193	28.68	0.8648	25.7M	34.03G
ShuffleFormer (Xiao et al., 2023)	33.87	0.9372	31.19	0.8949	27.85	0.8049	34.07	0.9516	31.76	0.9282	28.56	0.8718	50.61M	120.63G
GRL (Li et al., 2023)	33.83	0.9364	31.13	0.8932	27.75	0.8015	34.45	0.9521	32.18	0.9293	28.73	0.8685	3.49M	61.4G
CODE (Zhao et al., 2023)	33.69	0.9303	31.13	0.8861	27.98	0.7981	33.63	0.9442	31.24	0.9146	28.26	0.8528	12.23M	11.33G
AST (Zhou et al., 2024)	33.85	0.9324	31.23	0.8872	28.13	0.8032	33.93	0.9503	31.19	0.9185	27.85	0.8588	79.21M	38.47G
MambaIR (Guo et al., 2024)	33.91	0.9330	31.31	0.8903	28.10	0.8003	34.08	0.9517	31.78	0.9282	28.59	0.8719	3.17M	34.27G
VmambaIR (Shi et al., 2025)	33.98	0.9339	31.43	0.8921	28.23	0.8045	34.42	0.9496	32.12	0.9253	28.90	0.8721	15.09M	14.09G
Teacher(Ours)	34.16	0.9358	31.57	0.8943	28.34	0.8101	34.95	0.9531	32.66	0.9306	29.11	0.8785	47.27M	79.3G
MHKD(Ours)	33.89	0.9323	32.06	0.8921	28.16	0.8024	34.32	0.9487	32.01	0.9246	28.57	0.8602	716.37K	7.49G

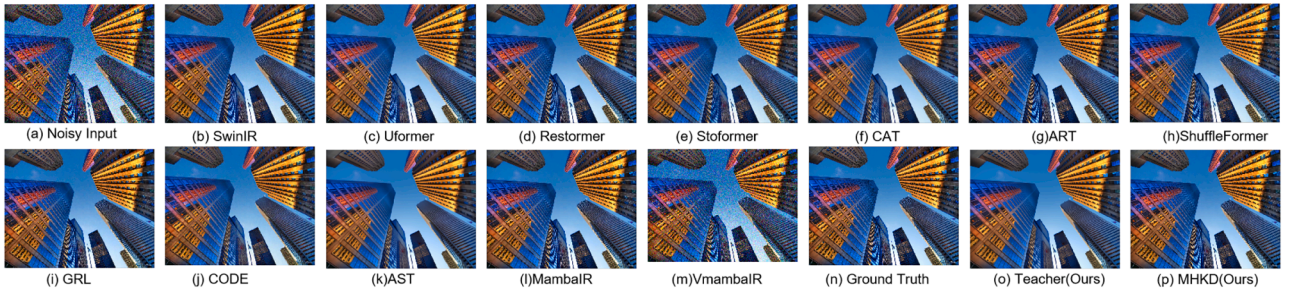


Fig. 4. Visual comparison results on image denoising task. Our MHKD student model preserves finer textures while removing noise, competing closely with larger transformer models.

Please zoom in on the figures for a better view.

Then, the loss function of MHKD is expressed as:

$$\mathcal{L}_{MHKD}^S = \mathcal{L}_{Bi}^S + \beta \mathcal{L}_{Filter}^S, \quad (10)$$

where, α , β are tunable parameters. Next, we construct the overall loss function as following:

$$\mathcal{L}_{Overall}^S = \mathcal{L}_{Rec}^S + \gamma \mathcal{L}_{Out}^S + \lambda \mathcal{L}_{MHKD}^S, \quad (11)$$

where, γ , λ are weight parameters. The Mamba-based student network is trained upon $\mathcal{L}_{Overall}^S$.

3. Experiments

3.1. Experimental setup

3.1.1. Datasets

Our goal is a task-specific yet widely applicable distillation framework: we train separate student models for each IR task (denoising, deraining, dehazing, deblurring) using the same Transformer teacher and MHKD procedure. This differs from an all-in-one model that handles multiple degradations with a single network. The experiments are conducted across numerous IR tasks in the single-one setting, including: 1) Image denoising uses the combination of BSD400 and WED dataset as training set and the testing is performed on BSD68 and Urban100. The noisy images are generated by adding white Gaussian noise on the clean images with corruption levels of $\sigma = 15, 25, 50$. 2) Image deraining chooses Rain200L, Rain200H (Yang et al., 2017), DDN-Data (Fu et al., 2017) as the benchmark datasets. Rain200L and Rain200H (Yang et al., 2017) contains 1800 training pairs and 200 test images with light and heavy rain streaks, respectively. DDN-Data (Fu et al., 2017) contains 12,600 pairs for training and 1400 ones for test. 3) Image dehazing is evaluated on SOTS (Li et al., 2018), O-HAZE (Ancuti et al., 2018),

I-HAZE (Ancuti et al., 2018). SOTS contains 500 outdoor hazy-clean image pairs. O-hazy (Ancuti et al., 2018) contains 45 real outdoor scenes using the same illumination parameters. I-hazy (Ancuti et al., 2018) contains 35 real image pairs of hazy and ground truth images. 4) Image deblurring picks GoPro (Nah et al., 2017), RealBlur-J (Rim et al., 2020), RealBlur-R (Rim et al., 2020) as the benchmark datasets. GoPro contains 2103 pairs for training and 1111 pairs for evaluation. Realblur-J and Realblur-R contains 3758 image pairs for training and 980 pairs for testing.

3.1.2. Architectural configuration

The number of HTB blocks in the teacher network are set to be [4, 6, 6, 8] from stage 1 to stage 4 in the encoder and decoder, respectively. And we set the head number of multi-head attention to be [2, 2, 4, 8] from stage 1 to stage 4 in the encoder and the decoder, respectively. The MLP expansion ratio in the HTB block is set as 4. In the Mamba-based student model, the number of RSSG is set to be 4, and the number of SSB is set to be 3. The FFM adopts an autoencoder structure, whose encoder and decoder are both composed of lightweight Transformer layers. FIM consists of a single Vision State Space Block (VSSB) and a tiny Hybrid Transformer Block (HTB), where the VSSB is responsible for mapping the teacher network features filtered by the FFM to the Mamba space. During the training phase, the FFM and FIM will generate a certain amount of computational overhead, which is specifically quantified as follows: the FFM has 89.2K parameters and a computational (FLOPs) overhead of 1.23G; the FIM has 102.5K parameters and a computational overhead of 1.57G. Together, they add an additional 191.7K parameters and 2.8G FLOPs overhead. Combined with the overall performance of the compressed student model (MHKD-Ours) in this paper, which only has 716.37K parameters and 7.49G FLOPs, it can be seen that the above

Table 2

Quantitative comparison of deraining performance on three benchmarks: Rain200L, Rain200H, and DDN-Data. The MHKD student model with only $\approx 716\text{K}$ parameters attains competitive performance, notably outperforming several larger models (e.g., SwinIR, CODE, VmambaIR) on Rain200L and closely matching others on Rain200H and DDN-Data.

Method	Rain200L		Rain200H		DDN-Data	
	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
SwinIR (Liang et al., 2021)	36.90	0.974	27.60	0.854	32.52	0.929
Uformer (Wang et al., 2022)	38.53	0.981	29.56	0.885	33.14	0.936
Restormer (Zamir et al., 2022)	40.93	0.988	30.22	0.901	34.35	0.947
Stoformer (Xiao et al., 2022)	38.39	0.982	28.47	0.868	33.30	0.936
CAT (Chen et al., 2022)	40.04	0.986	30.16	0.896	33.66	0.940
ART (Jiale et al., 2023)	40.11	0.986	30.36	0.900	33.55	0.938
ShuffleFormer (Xiao et al., 2023)	38.22	0.981	29.00	0.875	33.82	0.941
GRL (Li et al., 2023)	39.82	0.985	29.54	0.888	33.11	0.934
CODE (Zhao et al., 2023)	37.29	0.975	28.26	0.862	33.29	0.937
AST (Zhou et al., 2024)	35.04	0.964	27.96	0.846	33.55	0.938
MambaIR (Guo et al., 2024)	39.34	0.984	28.44	0.860	33.58	0.939
VmambaIR (Shi et al., 2025)	37.70	0.976	28.24	0.852	33.85	0.941
Teacher(Ours)	42.32	0.989	32.01	0.915	34.29	0.946
MHKD(Ours)	40.22	0.985	29.39	0.878	32.02	0.925

additional overhead does not significantly increase the overall training burden, but can bring obvious performance improvement.

3.1.3. Comparison methods and performance metrics

We compare our method with numerous state-of-the-art general purpose IR models, including SwinIR (Liang et al., 2021), Uformer (Wang et al., 2022), Restormer (Zamir et al., 2022), Stoformer (Xiao et al., 2022), CAT (Chen et al., 2022), ART (Jiale et al., 2023), ShuffleFormer (Xiao et al., 2023), CODE (Zhao et al., 2023), GRL (Li et al., 2023), AST (Zhou et al., 2024), MambaIR (Guo et al., 2024), VmambaIR (Shi et al., 2025). For a fair comparison, we re-implemented these baseline methods with the public codes and well-tuned parameters. The performance is measured by the popular metrics of PSNR and SSIM (Chen et al., 2024).

3.1.4. Training details

The model training and evaluation are performed on a single NVIDIA GeForce RTX 4090 graphics card. We use Adam with a momentum of $\beta_1 = 0.9$, $\beta_2 = 0.999$ and a weight decay of $5e-4$ as the optimizer. The initial learning rate is set to be 0.001, which is steadily decreased to $1e-6$ using the cosine annealing strategy. The maximal epoch number is set to be 1000 with a batch size of 8. The networks in all the methods are trained on patches at the size of 128×128 . The teacher model was trained from scratch by the authors using transformer blocks, rather than employing an existing pre-trained model. This custom design ensures architectural compatibility with the proposed distillation framework and provides a strong, dedicated knowledge source for the student network. The detailed training procedure and configuration are fully described in Section 2.2. The values of parameters are determined through a process of empirical tuning. We started with an initial guess with value of 1, and then performed random search, training multiple abbreviated versions of the model with different hyperparameter combinations to see which set yielded the best validation performance. Finally, the value of α , β , γ and λ are set to be 0.1, 0.1, 1, and 0.2.

3.2. Comparison with state-of-the-art methods

The quantitative comparison results on image denoising, image deraining, image dehazing and image deblurring tasks are shown in Tables 1–4, respectively. From the results, consistent observations can be obtained on all experimental tasks, i.e., the proposed teacher model behaves best with highest performance scores in both PSNR and SSIM,

illustrating that we can provide a powerful teacher network for the future in-depth research about image restoration model compression with offline knowledge distillation. Moreover, despite the performance gap between our student model and our teacher model, our student model has achieved comparable performance or even better performance than some other state-of-the-art models. Please note our student model embraces far fewer parameters and FLOPs. In particular, compared with our student model, Uformer (Wang et al., 2022) requires 73 times bigger parameters, and 12 times higher FLOPs, while our method outperforms it by a large margin up to 1.72 dB on SOTS-Indoor. The performance gains can also be observed on other datasets like Rain 200L and Gopro. The above analyses have obviously indicated the superiority of the proposed method to develop lightweight image restoration models.

The visualization comparison results on image denoising, image deraining, image dehazing and image deblurring tasks are shown in Figs. 4–7, respectively. From the results, we can see that our teacher model can yield clean images that have more natural appearance and better structural details, demonstrating the superiority of our teacher network. In addition, our student model also can generate very visually pleasing results. Notably, in the derained image shown in Fig. 5, our student model can remove the rain streaks better than some competing state-of-the-art models, while preserving finer structures and textures. In contrast, some comparison methods, such as Stoformer (Xiao et al., 2022), AST (Zhou et al., 2024) and VmambaIR (Shi et al., 2025), still have light rain streak residuals in the restored images. Moreover, considering our student requires the fewest parameters and FLOPs, the visualization results also have proved the proposed method has a great potential in IR community, especially for the resource-constrained realistic applications.

Please note that the compressed student model marginally outperforms its teacher in metrics such as PSNR or SSIM in certain configurations. This can be attributed to the distillation process acting as a form of structured regularization. While the teacher model possesses greater capacity, it may overfit to certain patterns or artifacts in the training data. In contrast, the student—constrained by its compact Mamba architecture—learns a more efficient and generalized representation through the bidirectional feature alignment enforced by MHKD. Specifically, the Feature Filter Module removes teacher-specific redundancies, and the Feature Interface Module encourages the student to develop features that are not only predictive but also robustly aligned in a shared latent space. Visual inspection of error maps further indicates that the student tends to produce smoother, more consistent reconstructions in homogeneous regions, whereas the teacher occasionally introduces high-frequency artifacts due to its self-attention mechanism. Thus, the distillation process not only compresses the model but can also refine its generalization, yielding a student that is both lighter and, in some cases, more accurate.

3.3. Evaluation on the real-world scenarios

In order to demonstrate the applications of our model in real-world scenarios or not, we conducted additional experiments on the RTTS dataset, which contains 4322 real-world hazy images from RESIDE. These real-world images are very challenging without ground-truth images. The visualization results on example images are shown in Fig. 8. From the results, we can see that the dehazing effect on RTTS is insufficient, characterized by residual haze and slight distortions. This suggests that the proposed method lacks adequate generalization when confronting the complexities of real-world imagery.

While real-world experiments demonstrate the practical applicability of our method, performance gains on real-world datasets are limited due to the complex and unpredictable real-world degradations that differ significantly from synthetic training data. This discrepancy arises because synthetic datasets cannot fully simulate uneven lighting, non-uniform haze, dense rain streaks, and complex motion blur encountered in real scenes, leading to incomplete restoration or minor artifacts. Fu-

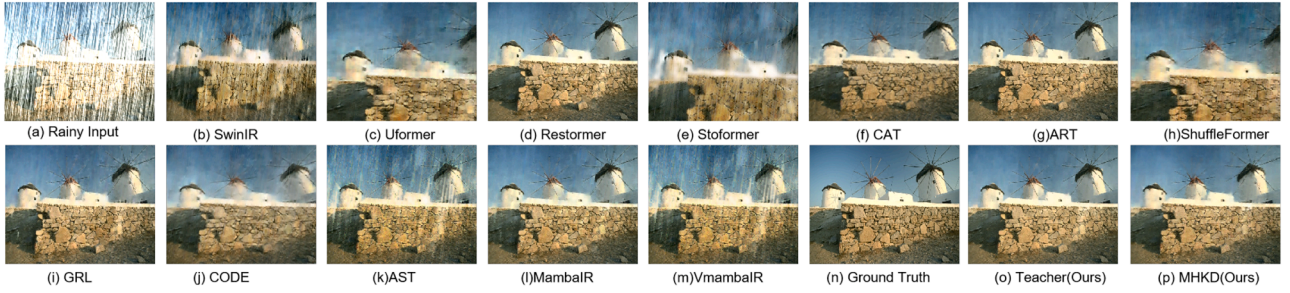


Fig. 5. Visual comparison results on image deraining task. Our MHKD student model effectively removes rain streaks while preserving structural details, outperforming methods such as Stoformer, AST, and VmambalR, which leave noticeable residual streaks or blur textures. The ground truth and input rainy image are shown for reference. Please zoom in on the figures for a better view.

Table 3

Quantitative comparison of dehazing performance (PSNR $\uparrow$ /SSIM $\uparrow$ /NIQE $\downarrow$) on SOTS-Indoor, O-HAZE, and I-HAZE datasets. The MHKD student model delivers competitive results—outperforming or matching several larger transformer and Mamba-based models (e.g., Uformer, CAT, VmambalR) while maintaining significantly lower computational cost.

Method	SOTS-Indoor			O-HAZE			I-HAZE		
	PSNR $\uparrow$	SSIM $\uparrow$	NIQE $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	NIQE $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	NIQE $\downarrow$
SwinIR (Liang et al., 2021)	33.71	0.982	3.237	21.78	0.694	4.152	22.69	0.850	4.525
Uformer (Wang et al., 2022)	32.06	0.980	3.485	21.24	0.641	4.126	20.09	0.769	5.053
Restormer (Zamir et al., 2022)	37.62	0.989	2.788	22.27	0.724	4.022	21.79	0.699	4.597
Stoformer (Xiao et al., 2022)	30.06	0.973	4.103	22.97	0.644	4.008	20.37	0.763	4.923
CAT (Chen et al., 2022)	35.12	0.985	3.073	21.14	0.655	4.468	18.68	0.714	5.224
ART (Jiale et al., 2023)	33.38	0.979	3.242	22.83	0.611	4.102	22.10	0.765	4.335
ShuffleFormer (Xiao et al., 2023)	30.64	0.968	3.9138	22.80	0.685	4.038	23.37	0.651	4.435
GRL (Li et al., 2023)	30.38	0.964	4.052	19.44	0.652	4.525	20.42	0.799	4.735
CODE (Zhao et al., 2023)	34.86	0.988	3.124	22.43	0.681	4.121	21.32	0.806	4.238
AST (Zhou et al., 2024)	32.42	0.980	3.826	21.73	0.688	4.124	20.43	0.740	4.913
MambalR (Guo et al., 2024)	33.04	0.981	3.335	23.76	0.752	3.836	18.57	0.834	4.359
VmambalR (Shi et al., 2025)	32.77	0.980	3.332	20.45	0.691	4.053	23.71	0.788	4.421
Teacher(Ours)	38.41	0.991	2.652	23.01	0.726	3.815	23.21	0.833	4.113
MHKD(Ours)	33.78	0.979	3.183	21.25	0.682	4.037	21.43	0.782	4.285

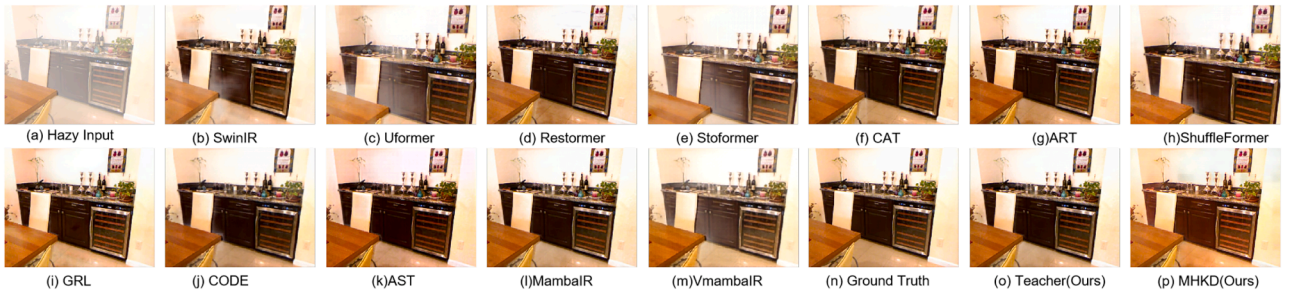


Fig. 6. Visual comparison results on image dehazing task. The MHKD student model achieves visually competitive results, preserving structural clarity while significantly reducing model size. Please zoom in on the figures for a better view.

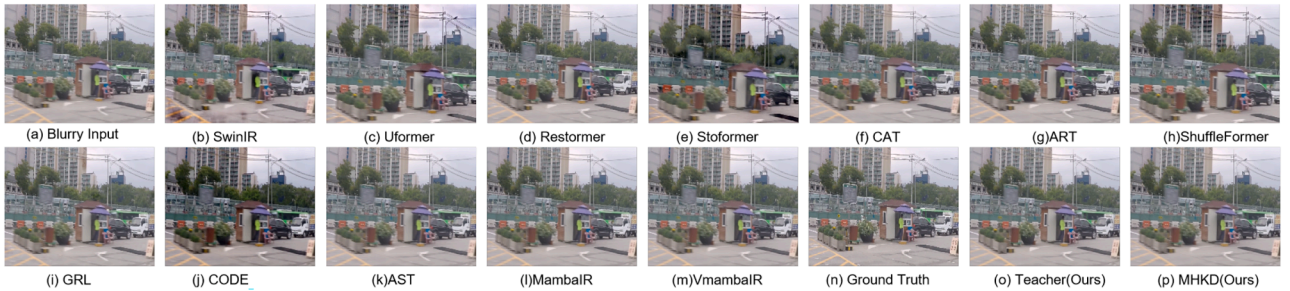


Fig. 7. Visual comparison results on image deblurring task. The MHKD student model delivers visually competitive results with significantly fewer parameters, effectively removing motion blur while preserving structural details. Please zoom in on the figures for a better view.



Fig. 8. Visual results on RTTS.

Table 4

Quantitative comparison of deblurring performance (PSNR↑/SSIM↑) on GoPro, RealBlur-J, and RealBlur-R datasets. The MHKD student model delivers competitive results, closely matching or surpassing several state-of-the-art models (e.g., SwinIR, CODE, AST) while requiring significantly fewer parameters and lower FLOPs.

Method	Gopro		Realblur-J		Realblur-R	
	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
SwinIR (Liang et al., 2021)	28.05	0.828	28.35	0.818	37.53	0.937
Uformer (Wang et al., 2022)	29.28	0.850	30.01	0.869	38.89	0.952
Restormer (Zamir et al., 2022)	30.78	0.888	30.20	0.877	39.33	0.957
Stoformer (Xiao et al., 2022)	29.61	0.859	29.88	0.870	39.03	0.953
CAT (Chen et al., 2022)	29.77	0.858	30.28	0.977	38.92	0.952
ART (Jiale et al., 2023)	28.80	0.836	30.02	0.871	39.46	0.957
ShuffleFormer (Xiao et al., 2023)	30.42	0.876	30.00	0.868	38.98	0.955
GRL (Li et al., 2023)	27.56	0.799	28.49	0.871	37.81	0.939
CODE (Zhao et al., 2023)	28.44	0.828	29.14	0.840	38.88	0.950
AST (Zhou et al., 2024)	27.70	0.807	29.24	0.842	37.74	0.937
MambaIR (Guo et al., 2024)	30.05	0.871	29.77	0.863	39.06	0.954
VmambaIR (Shi et al., 2025)	29.01	0.849	29.66	0.861	38.05	0.942
Teacher(Ours)	32.14	0.913	30.80	0.893	39.51	0.957
MHKD(Ours)	29.82	0.860	29.65	0.856	38.34	0.945

ture work will focus on improving real-world generalization by incorporating more diverse real-world degradation modeling and adaptive distillation strategies to narrow the gap between synthetic and real-image performance.

3.4. Computational efficiency comparison

Based on the experimental results presented in Table 5, the proposed MHKD method demonstrates significant advantages in runtime compared to state-of-the-art image restoration models. The student model, distilled via MHKD, contains only 0.71 million parameters and requires 7.49G FLOPs, which is substantially lower than most competing methods. For instance, Uformer has 52.66M parameters and 121.77 GFLOPs, while Restormer uses 26.88M parameters and 40.16 GFLOPs. Even lightweight designs like GRL (3.49M params, 61.4 GFLOPs) and VmambaIR (3.17M params, 34.27 GFLOPs) still lag behind in FLOPs efficiency. Regarding GPU memory consumption and runtime efficiency, the experiments utilize 128×128 three-channel images as input, with results averaged across 20 runs. As demonstrated in Table 5, the student model exhibits significant advantages, it achieves the lowest runtime, and although its GPU memory demand is second only to SwinIR, its execution speed substantially outperforms the latter.

3.5. Ablation studies

To show the effectiveness of MHKD, we conduct ablation studies on the overall loss function shown in Eq. (11). It is composed of $\mathcal{L}_{Rec}^S$, $\mathcal{L}_{Out}^S$, $\mathcal{L}_{MHKD}^S$, which represent the method without knowledge distillation, output-based knowledge distillation, and Mamba-oriented Heterogeneous Knowledge Distillation (MHKD), respectively. The experiments are conducted on the datasets of Rain200L, Rain200H, GoPro, and

Table 5

Computational efficiency comparison.

Method	Computational Efficiency			
	GPU(MB)	FPS(ms)	Param(M)	FLOPs(G)
SwinIR (Liang et al., 2021)	91	39.52	3.48	53.25
Uformer (Wang et al., 2022)	140	11.45	52.66	121.77
Restormer (Zamir et al., 2022)	241	18.12	26.88	40.16
Stoformer (Xiao et al., 2022)	234	21.09	50.47	123.97
CAT (Chen et al., 2022)	213	26.88	26.64	39.44
ART (Jiale et al., 2023)	151	19.45	25.7	34.03
ShuffleFormer (Xiao et al., 2023)	174	17.90	50.61	120.63
GRL (Li et al., 2023)	109	31.30	3.49	61.4
CODE (Zhao et al., 2023)	175	24.07	12.23	61.4
AST (Zhou et al., 2024)	186	9.02	79.21	38.47
MambaIR (Guo et al., 2024)	197	21.92	3.17	34.27
VmambaIR (Shi et al., 2025)	200	15.04	15.09	14.09
Teacher(Ours)	249	25.52	47.27	79.3
MHKD(Ours)	102	8.48	0.716	7.49

Realblur-J, and we report the results in Table 6. From the results, we can see that: (1) Compared with $\mathcal{L}_{Rec}^S$, the restoration performance of $\mathcal{L}_{Out}^S$ is significantly enhanced in both PSNR and SSIM across all the experimental datasets. In particular, the performance increments in PSNR happen most obviously on Rain 200L, where the gains amount to 1.58 dB. Also, the SSIM is boosted by up to 0.0173 and 0.0135 on datasets of Rain 200H and GoPro, respectively. The above observations have adequately suggested the proposed strategy of knowledge distillation-based image restoration compression is highly effective to improve the performance for the lightweight image restoration models. (2) Compared with $\mathcal{L}_{Out}^S$, the restoration performance is further increased by $\mathcal{L}_{MHKD}^S$ in both PSNR and SSIM across all the experimental datasets. The promising performance is mainly attributed to the fact that the proposed MHKD is an effective heterogeneous knowledge distillation method, which can successfully exchange information between transformer-based model and Mamba-based model. Additionally, the performance increments on GoPro and Realblur-J brought by $\mathcal{L}_{MHKD}^S$ are larger than the ones brought by $\mathcal{L}_{Out}^S$, illustrating that how to develop heterogeneous feature knowledge distillation is very significant for the IR model compression.

To rigorously evaluate the contribution of FFM, we conducted an ablation in which FFM was removed, retaining only the Feature Interface Module (FIM) alongside output-based distillation on Rain 200L, and we report the results in Table 7. The PSNR and SSIM on Rain200L are 40.15 and 0.984, respectively. Compared to the full MHKD, the Result shows a performance drop by 0.07 dB. This confirms that FFM plays a critical role in bridging the representational gap between Transformer and Mamba architectures, and that merely translating features via FIM without prior filtering yields suboptimal distillation. In addition, we also validated the bidirectional design of our FIM on Rain 200L, and we report the results in Table 8. We conducted an ablation comparing three distillation modes, i.e., $\mathcal{L}_{T \rightarrow S}^S$, $\mathcal{L}_{S \rightarrow T}^S$, $\mathcal{L}_{Bi}^S$. The Results demonstrate that the bidirectional variant $\mathcal{L}_{Bi}^S$ consistently outperforms both unidirectional settings, with gains of 0.2dB PSNR over $\mathcal{L}_{T \rightarrow S}^S$ and 0.4 dB over $\mathcal{L}_{S \rightarrow T}^S$ on average, confirming the value of transferring filtered teacher knowledge.

Impact of each parameter on PSNR

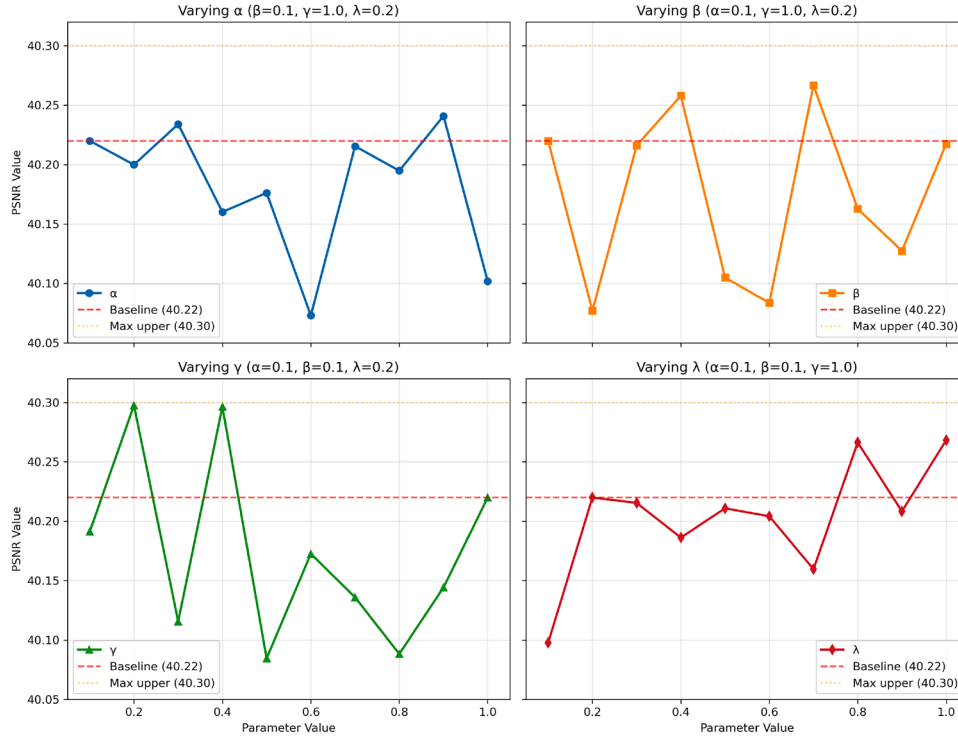


Fig. 9. Impact of each parameter on PSNR.

Table 6

The ablation results on deraining task and deblurring task. $\mathcal{L}_{Rec}^S$ represents the method without knowledge distillation. $\mathcal{L}_{Rec}^S + \mathcal{L}_{Out}^S$ represents output-based knowledge distillation. $\mathcal{L}_{Rec}^S + \mathcal{L}_{Out}^S + \mathcal{L}_{MHKD}^S$ represents the proposed MHKD.

Method	Rain200L		Rain200H		GoPro		Realblur-J		#Param↓	#Flops↓
	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑		
$\mathcal{L}_{Rec}^S$	38.51	0.978	28.36	0.857	28.92	0.838	29.08	0.839	716.37K	7.49G
$\mathcal{L}_{Rec}^S + \mathcal{L}_{Out}^S$	40.09(+1.58)	0.984	29.06(+0.70)	0.875	29.32(+0.40)	0.852	29.30(+0.22)	0.847		
$\mathcal{L}_{Rec}^S + \mathcal{L}_{Out}^S + \mathcal{L}_{MHKD}^S$	40.22(+0.13)	0.985	29.39(+0.33)	0.878	29.82(+0.50)	0.860	29.65(+0.35)	0.856		

Table 7

Ablation results on Rain200L dataset. w/o FFM represents the method without FFM.

Method	Rain200L	
	PSNR ↑	SSIM ↑
full MHKD	40.22	0.985
w/o FFM	40.15 (-0.07)	0.984 (-0.0006)

Table 8

Ablation comparing three distillation modes on Rain200L dataset.

Method	Rain200L	
	PSNR ↑	SSIM ↑
$\mathcal{L}_{T \rightarrow S}^S$	40.02	0.982
$\mathcal{L}_{S \rightarrow T}^S$	39.82	0.979
$\mathcal{L}_{Bi}^S$	40.22	0.985

3.6. Parameter analysis

In this section, we analyzed the influence of the parameter on the datasets of Rain200L. We varied each hyperparameter individually over a range of values while keeping the others fixed at our default settings.

The results are shown in Fig. 9. From the results, we can see that PSNR fluctuates within a narrow band of ± 0.15 dB across all tested configurations. This indicates that MHKD is not overly sensitive to the exact values of these weights and that the chosen configuration lies in a broad plateau of good performance.

4. Conclusion

This work strives to arrive at a lightweight yet fast generic image restoration model. To this end, we formally propose a strategy of knowledge distillation-based image restoration model compression, which is formulated as an off-line knowledge distillation. Specifically, a cumbersome yet high-quality teacher network is built upon the transformer and pre-trained with the reconstruction loss, while a lightweight student network is established with the Mamba. The model compression is implemented on the student network with the proposed Mamba-oriented Heterogeneous Knowledge Distillation method (MHKD). MHKD can guarantee an efficient knowledge shift between the heterogeneous architectures of transformer and Mamba. Extensive experiments on multiple image restoration tasks have apparently shown our method can bring a significant performance boost for the lightweight model. Nevertheless, the empirical validation in this paper is limited to the Transformer-to-Mamba heterogeneous pair. Generalization to other architecture combinations (e.g., CNN $\rightarrow$ Mamba, Transformer $\rightarrow$ RNN, or Mamba $\rightarrow$ Trans-

former) has not been experimentally demonstrated and remains an open direction for future work. Readers should therefore interpret our claims of "general applicability" as referring to the flexibility of the proposed formulation, not as proven performance across all heterogeneous pairs.

Yet, the proposed method still suffers the following limitations: (1) The method has only been validated for the Transformer-to-Mamba distillation pair. Its effectiveness for other heterogeneous architecture pairs (e.g., CNN→Mamba, Transformer→SSM variants) is not yet established and requires further investigation. (2) Due to the compact size, the lightweight model is still inferior to some state-of-the-art models in big size on certain complex tasks, indicating a fundamental trade-off between model size and restoration fidelity for severe degradations. (3) The distillation process itself introduces computational overhead due to the additional Feature Filter Module (FFM) and Feature Interface Module (FIM), and the optimization function needs careful tuning of hyperparameters, which can be complex and dataset-dependent. In addition, typical failure cases include residual rain streaks in dense, overlapping precipitation, incomplete haze removal in non-uniform lighting, and over-smoothed textures under extreme noise, highlighting the student's limited capacity for highly ill-posed degradations. Therefore, future work will explore the integration of other emerging efficient architectures beyond Mamba, which could provide significant leaps in achieving lightweight yet powerful image restoration models. Moreover, we will also focus on developing more efficient and adaptive distillation strategies in future, such as leveraging dynamic or neural architecture search-based methods to automate the knowledge transfer process.

CRedit authorship contribution statement

Sai Yang: Writing – review & editing, Writing – original draft, Data curation; **Bin Hu:** Writing – review & editing, Writing – original draft, Resources, Project administration; **Xiaoxin Wu:** Resources; **Fan Liu:** Resources; **Wanzhi Wen:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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